

Considerations in Living Kidney Donation

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CHAPTER OUTLINE

EVALUATION AND SELECTION OF THE LIVING KIDNEY DONOR CANDIDATE, 2051
 PAIRED DONATION AND INCOMPATIBLE LIVING DONOR TRANSPLANTS, 2059
 DONOR NEPHRECTOMY APPROACHES, 2060

LIVING KIDNEY DONOR FOLLOW-UP, 2060
 RISKS AND OUTCOMES OF LIVING KIDNEY DONATION, 2061
 ETHICAL, LEGAL, AND POLICY CONSIDERATIONS, 2065

KEY POINTS

- Considering practicality, test availability, and costs, the initial estimated glomerular filtration rate (eGFR) should be based on serum creatinine (eGFR_{cr}) and confirmation using one or more of the following measurements according to their availability: measured GFR from clearance of exogenous radiolabeled filtration markers, measured creatinine clearance based on collecting a timed (24-hour) urine specimen, eGFR based on serum creatinine and cystatin, or repeated eGFR_{cr}, with the latter being the least preferred approach.
- Using the 2021 race-free Chronic Kidney Disease Epidemiology Collaboration equation may result in more Black kidney donor candidates being excluded from donation in the screening phase, emphasizing the need for confirmatory testing in candidates with borderline eGFR_{cr}. Genetic testing may be offered to asymptomatic donor candidates with a family history of genetic disease after appropriate counseling that includes the challenges, limitations, and uncertainties inherent in the process and results.
- The presence of two *APOL1* renal risk variants increases the lifetime chance of developing kidney failure, but the effects of kidney donation on this risk are unknown. Refining understanding of *APOL1*-related risk for donor candidates is a topic of active research.
- There is considerable variability in the practice patterns for accepting living kidney donors with a history of COVID-19 infections, indicating the need for further research to guide optimal practices to ensure the safety of living donation and living donor kidney transplantation as COVID-19 transitions from a pandemic to an endemic infection, as well as to guide future pandemic responses.

For eligible patients with end-stage kidney failure (ESKF), living donor kidney transplantation (LDKT) is the preferred treatment option due to superior outcomes for the recipient. Compared with deceased donor kidney transplantation (DDKT), LDKT is associated with improved patient and graft survival. Five-year patient and graft survival are higher for LDKT recipients than DDKT recipients (patient survival: 93% vs. 85%; graft survival: 88% vs. 78%).¹ Other benefits for the recipient include the potential for preemptive transplantation or shorter waiting time until transplant, allowing recipients to avoid or decrease the risks associated with chronic dialysis.^{2,3}

Further, LDKT can be scheduled with advance notice at a time that works well for donors and recipients, compared with the urgent nature of DDKT. This may allow for better optimization of pretransplant conditions before surgery. LDKT may provide additional opportunities for better genetic matching and a lower risk of rejection. Lastly, with LDKT, the cold ischemia time is shortened and the risk of delayed graft function is decreased. The practice of LDKT is based on the principle that these benefits for the recipient outweigh the generally low risks to the donor.⁴ The risks associated with living donor nephrectomy are outlined in this chapter.

Every year, more than 35,000 LDKTs are performed worldwide, representing almost 40% of all kidney transplant surgeries.⁵ Rates of LDKT vary across nations with the highest rates, per million population (pmp) in 2023, in countries such as Turkey (37 pmp), Saudi Arabia (34 pmp), and Israel (32 pmp) with somewhat lower rates in the United States (19 pmp) and Canada (13 pmp).⁶ Thus considerations for the safe, effective, and ethical practices in living kidney donation have global significance, affecting more than 90 countries worldwide.

EVALUATION AND SELECTION OF THE LIVING KIDNEY DONOR CANDIDATE

Living kidney donors must undergo a thorough evaluation and selection process to ensure that the short- and long-term risks to the donor are minimized (Fig. 70.1). The living donor evaluation process should also be completed in a timely fashion to improve the donor candidate experience and optimize recipient outcomes. One international, multicentered study reported that the median time from the beginning of the donor candidate evaluation until donor nephrectomy was 10 months.⁷ Avoidable delays can lead to dialysis initiation for preemptive transplant candidates, lengthen dialysis time for those patients on dialysis, and reduce donor satisfaction.³ Opportunities to improve efficiency include the choice, timing, and sequence of evaluation tests and visits; the number of candidates assessed at the same time; the use of navigators; and monitoring of evaluation timeliness as a quality metric.^{8,9}

Multiple guidelines assist clinicians in the complex process of donor evaluation and selection (Table 70.1). A systematic review of these clinical practice guidelines found that while many recommendations were consistent, important variations exist and many appeared to lack methodological rigor.¹⁰ The 2017 Kidney Disease: Improving Global Outcomes (KDIGO) “Clinical Practice Guideline on the Evaluation and Care of Living Kidney Donors” provides a comprehensive set of best practice recommendations based on a systematic evidence review, de novo evidence generation, and expert opinion when evidence is lacking.¹¹ When possible, the guideline recommends that transplant programs establish numeric thresholds for short- and long-term postdonation risks, above which the program will not accept the candidate for donation (Fig. 70.2). The guideline also demonstrates how tools can be developed to help estimate a donor candidate’s risk of long-term complications, such as ESKF, based on their individualized set of predonation demographic and health characteristics.¹²

The evaluation process should include a comprehensive history, physical examination, laboratory and radiologic investigations, and specialist consultations (Table 70.2). Aspects of the process may vary by region and transplant center, including the order and timing of the components and what is considered required or additional testing. The 2017 KDIGO living kidney donor guideline recommends that all donor candidates should be evaluated using the same criteria, regardless of who the intended recipient is.¹¹ Table 70.3 summarizes several contraindications to living kidney donation.

KIDNEY FUNCTION

The purpose of evaluating glomerular filtration rate (GFR) in kidney donor candidates is to detect kidney disease and to project long-term outcomes for the candidate and their recipient should they go on to donation. Recommended methods for evaluating GFR in donor candidates are based on the 2012 KDIGO Chronic Kidney Disease (CKD) guideline.^{13,14} Considering practicality, test availability, and costs, the 2017 KDIGO living donor guideline recommends initial estimated GFR (eGFR) based on serum creatinine (eGFRcr) and confirmation using one or more of the following measurements according to their availability: measured GFR (mGFR) from clearance of exogenous radiolabeled filtration markers, measured creatinine clearance (mCrCl) based on collecting a timed (24-hour) urine specimen, eGFR based on serum creatinine and cystatin (eGFRcr-cys), or repeated eGFRcr; the latter being the least preferred approach.^{11,13,14}

The implication of inclusion of a race coefficient in commonly used equations to estimate GFR is a controversial topic due to concerns related to inaccuracy leading to health care disparities. A National Kidney Foundation (NKF)/American Society of Nephrology (ASN) Task Force assessed the continued use of race in GFR calculation and recommended implementation of the CKD-EPI (CKD Epidemiology Collaboration) 2021 equation without the race variable.¹⁵ At the higher eGFR thresholds recommended by KDIGO for donor candidacy, more Black adults may be declined early based on initial eGFR screening if the CKD-EPI equation without the race variable is used. In one study, if the eGFR was calculated using the CKD-EPI equation without the race variable, the proportion of kidney donor candidates that may be declined on the basis of acceptable eGFR thresholds would change from 38.5% to 40.6%, newly disqualifying 2.1% of Black adults from kidney donation.¹⁶ Thus confirmatory testing with mGFR or mCrCl is required in the United States, and implications of using race-neutral eGFR for donor screening should be carefully considered.

One strategy to optimize efficiency may include obtaining eGFRcr and eGFRcr-cys before a donor candidate’s visit to the center to determine the posttest probability that mGFR is $<60 \text{ mL/min/1.73 m}^2$. Donor candidates with a high posttest probability of $\text{mGFR} < 60 \text{ mL/min/1.73 m}^2$ could be excluded rather than undergo a complete evaluation (Fig. 70.3).^{17,18} If there is a high probability that the mGFR is greater than a threshold needed for donation, and if urine albumin-to-creatinine ratio (ACR) is low, then these tests could be repeated for confirmation without mGFR, mCrCl, or timed albumin excretion rate (AER). Alternatively, donor candidates with either a high probability of $\text{mGFR} < 60 \text{ mL/min/1.73 m}^2$ or high urine ACR could be reliably excluded from donation, with any additional testing conducted as indicated for the purposes of supporting the candidate’s long-term health. Donor candidates with eGFR or urine ACR in intermediate ranges would require confirmatory tests with mGFR, mCrCl, and/or urine AER.

Nephrectomy in a healthy donor (removal of 50% of the nephron mass) is associated with an adaptive increase of the remaining kidney’s GFR such that postdonation GFR is stable at approximately 60% to 70% of the predonation

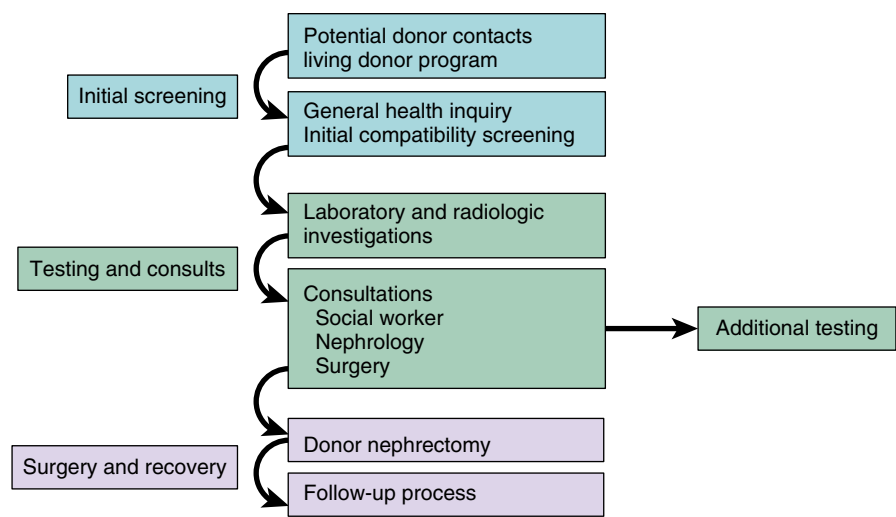


Fig. 70.1 Overview of the living kidney donor evaluation process.

Table 70.1 Existing Guidelines and Policies on Evaluation of Living Kidney Donors

Guidelines	Year	National/International
Organ Procurement & Transplantation Network (OPTN) Policy 14: Living Donation ⁶⁰	2024	United States
British Transplantation Society (BTS): United Kingdom Guidelines for Living Donor Kidney Transplantation ⁹⁹	2018	United Kingdom
Kidney Disease: Improving Global Outcomes (KDIGO): Evaluation and Care of Living Kidney Donors ¹¹	2017	International
Kidney Paired Donation (KPD) Protocol for Participating Donors ⁴⁵	2015	Canada
European Renal Best Practice (ERBP) Guideline on Kidney Donor and Recipient Evaluation and Perioperative Care ¹⁰⁰	2013	Europe
Caring for Australasians with Renal Impairment (CARI) ¹⁰¹	2010 (archived)	Australia/New Zealand

value over the first-decade postdonation (Fig. 70.4),¹⁹ although the degree of adaption can vary (e.g., may be lower in donors who are older, obese, and hypertensive).²⁰ This information is relevant in considering an acceptable baseline predonation GFR. The 2017 KDIGO living kidney donor guideline recommends a GFR ≥ 90 mL/min/1.73 m² as an acceptable level of kidney function for donation, while donor candidates with GFR < 60 mL/min/1.73 m² should not donate.¹¹ The decision to approve donor candidates with GFR 60 to 89 mL/min/1.73 m² should be individualized on the basis of demographic and health profiles. It is important to note that some other guidelines on the assessment of living kidney donor candidates incorporate an age-specific criteria for donor selection based on GFR.²¹ There are concerns that more candidates may be considered potentially ineligible when an absolute GFR threshold of 90 mL/min/1.73 m² is used compared with age-adapted GFR thresholds, and this may be more pronounced with eGFR measurements compared with mGFR measurements.²² Whether an absolute threshold or an age-adapted GFR threshold is used, transplant centers should consider an overall personalized risk estimate to determine donor eligibility.¹⁸

ALBUMINURIA

Elevated protein in the urine (proteinuria) may suggest the presence or risk of developing kidney disease due to increased permeability of the glomeruli to protein and/or an inability of the renal tubules to reabsorb protein. Until acceptable standardization methods are available for quantifying deficiencies in tubular reabsorption, urine albumin remains the most reliable indicator of kidney disease, standardized to urinary creatinine as the ACR. The 2017 KDIGO living kidney donor guideline recommends initial evaluation using urine ACR in a random urine specimen with confirmation by AER (from a timed urine specimen) or otherwise a second random urine ACR.¹¹ Donor candidates with an AER > 100 mg/day should not donate. Such candidates have moderately increased albuminuria and are at an elevated risk of developing CKD in their lifetime. Candidates with an AER < 30 mg/day may be acceptable for donation, while the decision to approve donor candidates with AER 30 to 100 mg/day should be individualized on the basis of demographic and health profiles in relation to the transplant program's acceptable risk threshold.

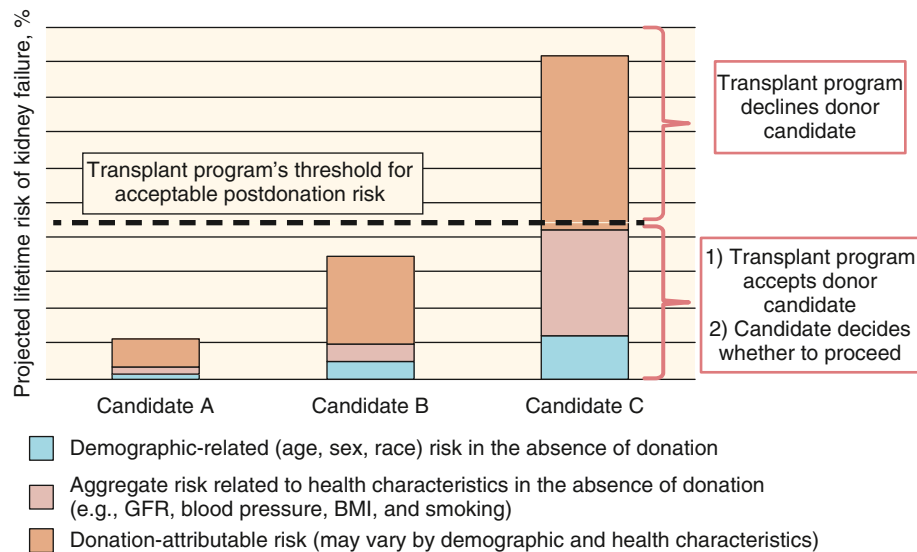


Fig. 70.2 KDIGO framework to accept or decline donor candidates based on a transplant program's threshold of acceptable projected lifetime risk of kidney failure, quantified as the aggregate of risk related to demographic and health profile and donation-attributable risks. BMI, Body mass index; GFR, glomerular filtration rate; KDIGO, Kidney Disease: Improving Global Outcomes. (Reproduced with permission.¹¹)

Table 70.2 Components of the Living Kidney Donor Evaluation

Medical/Surgical History	Examples	Medical/Surgical History	Examples
Genitourinary diseases	Hematuria Proteinuria Kidney stones	Infections	Urinary tract infections Hepatitis Human immunodeficiency virus Syphilis Sexually transmitted infections Severe acute respiratory syndrome Meningitis/Encephalitis Creutzfeldt-Jakob disease Tuberculosis COVID-19 (Coronavirus disease 2019)
Cardiac diseases	Coronary artery disease Myocardial infarction Heart failure Arrhythmia	Cancer	Melanoma Renal cell carcinoma Breast cancer Lung cancer Colon cancer Hematologic cancer (leukemia, lymphoma) Monoclonal gammopathy
Cardiac risk factors	Hypertension Impaired fasting glucose Impaired glucose tolerance Diabetes mellitus Hyperlipidemia	Pregnancy-related complications	Gestational hypertension Gestational diabetes Preeclampsia/Eclampsia Future childbearing plans
Peripheral vascular disease	Intermittent claudication	Allergies/Medications	
Pulmonary diseases	Asthma Chronic obstructive pulmonary disease Obstructive sleep apnea	Allergies	Anesthetic reactions
Gastrointestinal diseases	Peptic ulcer disease Inflammatory bowel disease	Nephrotoxic medications	Lithium
Hematologic diseases	Deep vein thrombosis Pulmonary embolism Bleeding disorders Blood transfusions	Chronic pain medications	Nonsteroidal antiinflammatory drugs Opioids
Neurologic diseases	Stroke		
Autoimmune diseases	Systemic lupus erythematosus Rheumatoid arthritis		
Rheumatologic diseases	Gout		
Psychiatric conditions	Anxiety Depression Suicide attempts		

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Table 70.2 Components of the Living Kidney Donor Evaluation (Continued)

Medical/Surgical History		Medical/Surgical History	
Examples		Examples	
Family History		Renal	Serum creatinine with estimation of glomerular filtration rate
Kidney diseases	Chronic or end-stage kidney failure (attempt to clarify etiology) Polycystic kidney disease Alport syndrome		Urea
Genitourinary diseases	Congenital anomalies of kidney and urinary tract Kidney stones		Urinalysis
Cardiac diseases	Sudden cardiac death Early coronary artery disease		Urine microscopy
Cardiac risk factors	Hypertension Diabetes mellitus		24-hour urine (creatinine clearance, protein excretion, metabolic stone panel including calcium, oxalate, uric acid, citric acid, sodium) ^a
Cancer	Renal cell carcinoma Breast cancer Colon cancer		Protein-creatinine ratio
Social History			Albumin-creatinine ratio
Smoking, alcohol, drug use	Abuse/Dependency Intravenous drug use		Complete blood count (hemoglobin, white blood cells, platelets)
Employment, dependent care responsibilities	Health care workers Veterinarians Dependent care responsibilities that may be interrupted by donation		International normalized ratio
Health insurance status			Partial thromboplastin time
Financial barriers to living donation		Electrolytes (sodium, potassium, bicarbonate, calcium, phosphate, magnesium)	
Social supports	Marital status Living arrangements	Fasting lipid profile (total, high-density lipoprotein, low-density lipoprotein cholesterol, triglycerides)	
Exposure history	Sexual history Tattoos Body piercings Travel or prolonged residency Country of origin/birth Incarceration	Pancreas enzymes (lipase, amylase)	
Exercise tolerance		Liver enzymes (aspartate aminotransferase, alanine aminotransferase, alkaline phosphatase, gamma-glutamyl transferase)	
Physical Examination		Diabetes mellitus	Bilirubin
Vital signs	Blood pressure Heart rate and rhythm Body mass index based on height and weight Waist circumference		Albumin
Head and neck			Uric acid
Cardiac examination			Thyroid function test ^a
Respiratory examination			β-human chorionic gonadotropin ^a
Abdominal examination			Apolipoprotein L1 (APOL1) genotyping ^a
Skin examination	Lymphadenopathy Mucocutaneous lesions Needle tracks		Genetic kidney disease panel ^a
Laboratory Investigations			Fasting blood glucose
Compatibility	ABO blood group (including subtype A1, if indicated) Human leukocyte antigen typing Crossmatch Eplet matching (e.g., in some paired exchange programs) ^a		Hemoglobin A1C
			Oral glucose tolerance test ^a
		Urine culture and sensitivity	
		Cytomegalovirus	
		Epstein-Barr virus	
		Hepatitis B (Hepatitis B surface antigen, Hepatitis B core antibody)	
		Hepatitis C antibody	
		Human immunodeficiency virus antibody	
		Human T-lymphotropic virus	
		Herpes simplex virus	
		Varicella zoster virus	
		Treponema pallidum (syphilis)	
		Mycobacterium tuberculosis (purified protein derivative, interferon-γ release assay)	
		Malaria ^a	

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Table 70.2 Components of the Living Kidney Donor Evaluation (Continued)

Medical/Surgical History	Examples	Medical/Surgical History	Examples
	<i>Trypanosome cruzi</i> ^a Schistosomiasis ^a Human herpes virus ^a Strongyloides ^a Typhoid ^a Brucellosis ^a West Nile virus ^a Zika virus ^a Chagas ^a Toxoplasmosis ^a COVID-19 (Coronavirus disease 2019) Cancer screening Urine cytology ^a Fecal occult blood test or fecal immunochemical test ^a Prostate-specific antigen ^a Pap smear ^a	Renal	Abdominal ultrasound Abdominal computed tomography or magnetic resonance angiography Measured glomerular filtration rate (iothalamate, ethylenediaminetetraacetic acid, diethylenetriaminepentaacetic acid) Cystoscopy ^a Renal biopsy ^a
		Cardiac	Electrocardiogram 24-hour Holter monitor ^a 24-hour ambulatory blood pressure monitor ^a Echocardiography ^a Stress echocardiography ^a Exercise stress test ^a Coronary angiogram ^a
		Pulmonary	Chest radiograph Pulmonary function test ^a
		Cancer screening	Mammogram ^a
Radiologic Investigations			

HEMATURIA

The persistent presence of blood in the urine (hematuria) is another indicator for the presence or risk of developing kidney disease. Presence of hematuria is established by visualizing two to five red blood cells per high-powered field on microscopic evaluation. “Persistence” is established if hematuria is observed in more than 50% of urine samples obtained from two to three separate occasions. When hematuria is persistent, further investigation is warranted, which may include urine culture for bacterial or fungal infection (this may be treated without affecting candidacy); 24-hour urine stone panel; cystoscopy; imaging to rule out a urinary tract malignancy or asymptomatic stone (imaging, such as a computed tomography angiogram, is often done as part of the donor evaluation regardless of the presence of hematuria); genetic testing donor candidates with a family history of kidney disease (see next section on Genetic Kidney Diseases); and a kidney biopsy to rule out underlying kidney disease (thin basement membrane may not be a contraindication to donation).

GENETIC KIDNEY DISEASES

If the donor candidate is biologically related to the intended recipient, the cause of the recipients’ kidney disease should be well understood before accepting the candidate. Candidates

with a genetic kidney disease are not eligible, generally, to become donors. If a candidate has a family history of a genetic kidney disease, the candidate may be eligible to donate if the risk of developing kidney disease after donation is acceptably low and the risks are discussed with the candidate. Genetic diseases that may be assessed during the donor candidate evaluation include autosomal dominant polycystic kidney disease, atypical hemolytic uremic syndrome, Alport syndrome, Fabry disease, familial focal segmental glomerulosclerosis, and autosomal dominant tubulointerstitial kidney disease. Genetic testing may be offered to asymptomatic donor candidates with a family history of genetic disease after appropriate counseling that includes the challenges, limitations, and uncertainties inherent in the process and results. One approach to genetic testing of donor candidates with a family history of genetic disease involves sequential testing of the affected family member first (often this is the intended recipient candidate) to determine the genetic basis of the disease (and to see if it can be identified with modern testing) and then do targeted testing of the donor candidate (Fig. 70.5).²³

For donor candidates of sub-Saharan African ancestry, testing for apolipoprotein L1 (*APOL1*)-related kidney disease risk alleles may be offered.²³ The presence of two *APOL1* risk alleles increases the lifetime chance of developing kidney failure,²⁴ yet the effect of kidney donation on this risk is

Table 70.3 Contraindications to Living Kidney Donation^{11,60}

Absolute Contraindications

Both <18 years old and mentally incapable of making an informed decision
High suspicion of donor coercion
High suspicion of illegal financial exchange between donor and recipient
ABO or HLA (human leukocyte antigen) incompatibility without a planned management protocol
Impaired kidney function (e.g., glomerular filtration rate <60 mL/min/1.73 m²)
Macroalbuminuria (>300 mg/day)
Hematuria with evidence of kidney disease
Diagnosis of IgA nephropathy or Alport syndrome
Bilateral/Recurrent kidney stones
Uncontrolled hypertension or hypertension with evidence of end-organ damage
Type I diabetes mellitus
Acute/Chronic bacterial or viral infection (Hepatitis B, West Nile virus)
Active *Mycobacterium tuberculosis* or urinary tuberculosis
Untreated or incompletely treated syphilis infection
Active cancer or incompletely treated cancer
History of melanoma, testicular cancer, choriocarcinoma, hematological cancer, monoclonal gammopathy, bronchial cancer, metastatic cancer
Treatment with anticoagulation without a planned management protocol
Current pregnancy
Uncontrolled psychiatric illness
Active substance or alcohol abuse disorder

Relative Contraindications

Obesity (e.g., body mass index >35 kg/m² or >30 kg/m² with other comorbidities)
Microalbuminuria (>30 mg/day)
Two apolipoprotein L1 (*APOL1*) renal risk alleles
Family history of genetic kidney disease with identification of pathogenic variants in the donor candidate
Chronic illness (cardiac, pulmonary, gastrointestinal, neurologic)
Type II diabetes mellitus, prediabetes, or metabolic syndrome
Premenopausal women with a history of gestational diabetes
Premenopausal women with a history of toxemia in pregnancy
Sickle cell trait
Hepatitis C or human immunodeficiency virus without a planned management protocol or clinical trial¹⁰³

unknown but is a topic of active research.²⁵ One study from the United States compared 136 Black living kidney donors, of whom 19 were considered *APOL1* high-risk genotype (two risk alleles) and 117 were considered *APOL1* low-risk genotype (zero or one allele).²⁶ Donors who were *APOL1* high-risk genotype had a lower predonation eGFR compared with donors who were *APOL1* low-risk genotype (98 vs. 108 mL/min/1.73 m², *P* = 0.03) and lower postdonation eGFR (57 vs. 67 mL/min/1.73 m², *P* = 0.02) after a median of 12 years postdonation and after adjusting for predonation eGFR. The rate of decline was also faster in donors who were *APOL1* high-risk genotype (1.1 vs. 0.4 mL/min/1.73 m² per year, *P* = 0.02), although this was not significantly different from matched nondonor controls. Some studies

have suggested the potential importance of genotyping for a modifying genetic variant in the *APOL1* gene encoding an amino acid change N264K, which confers protection when coinherited with high-risk genotypes.²⁷ This modifying N264K variant is uncommon and may be inherited with G2 but is mutually exclusive with G1 renal risk variants. Future research should assess whether this variant increases the safety for kidney donation in such potential donors.

KIDNEY STONES

A renal calculus in the donor's remaining kidney may affect kidney function if it results in ureteral obstruction. Reassuringly, living kidney donors without a history of predonation kidney stones do not appear to have an increased risk of postdonation kidney stones requiring treatment with surgical intervention compared with healthy, matched nondonor controls (median follow-up of 8 years).²⁸ Evaluation of kidney stones in living kidney donor candidates includes a history from the candidate, laboratory investigations including urine samples for evaluation of persistent microscopic hematuria, and renal imaging such as computed tomography. If suspected, further investigations may be performed, including parathyroid hormone measurements and 24-hour urine collections for metabolic testing. A history of previous kidney stone disease does not necessarily rule out donation, particularly small, unilateral, nonrecurrent stones.¹¹ There is also the option to remove small kidney stones at the time of procurement before transplantation.²⁹

BLOOD PRESSURE

Sustained elevated blood pressure is a common cause of kidney disease, and conversely, kidney disease may accelerate the development of high blood pressure. Candidates with hypertension are eligible for donation only if their blood pressure can be well controlled with a simple regimen (commonly defined as one or two antihypertensive medications), and they are without end-organ damage related to their hypertension.¹¹ The characteristics used to disqualify a candidate from donation, such as the level of systolic and/or diastolic blood pressure, nature of the antihypertensive medications used (e.g., number of agents, class of drugs, and dosage used), and other candidate characteristics (e.g., hypertension onset at a young age) may vary across programs. Trained personnel should perform blood pressure measurements on at least two separate occasions. An ambulatory (e.g., 24-hour) blood pressure monitor may be used if hypertension is suspected. In a single-center study of 578 prospective donors in the United States, lower thresholds of <123/82 mm Hg and <120/78 mm Hg for office and automated blood pressure used to prompt 24-hour ambulatory blood pressure monitor recording improved the sensitivity (from 79% to 87%) of diagnosing hypertension (based on a threshold of <135/85 mm Hg by daytime ambulatory blood pressure monitoring) and reduced the prevalence of missed (masked) hypertension by 2%.³⁰ Thus using lower screening thresholds may identify donor candidates who would most benefit from ambulatory blood pressure monitoring to diagnose hypertension.

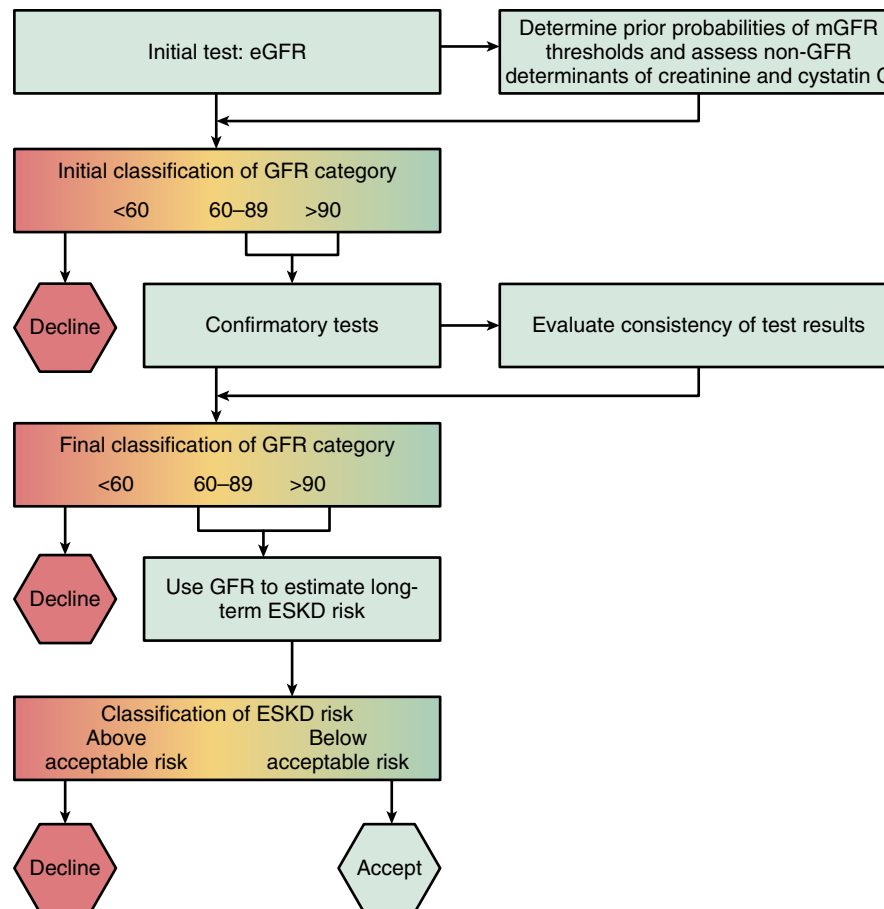


Fig. 70.3 Assessment of glomerular filtration rate. Initial test. eGFRcr is the initial test for most candidates. eGFRcys may be the preferred initial test for candidates with variations in non-GFR determinants of serum creatinine (e.g., variation in muscle mass or diet). Interpretation of eGFR should include consideration of the probability that mGFR is above or below threshold for decision making. High likelihood that eGFR <60 mL/min/1.73 m² is justification for a decision to decline without further consideration. **Confirmatory tests.** mGFR or mCrCl are required in the United States. Elsewhere, eGFRcr-cys can be acceptable if mGFR or mCrCl is not available and eGFRcys was not used as the initial test. Repeat eGFRcr can be acceptable if none of the other confirmatory tests are available but are not preferred. Inconsistent test results suggest inaccuracy of one or more tests, which should be discarded or repeated. Very high likelihood that mGFR <60 mL/min/1.73 m² is justification for a decision to decline without further consideration.

Using GFR to estimate long-term ESKF risk. Long-term estimated risk of ESKF is compared with the transplant center threshold for acceptable risk. Long-term risk in the absence of donation can be computed from demographic and clinical characteristics including GFR (<http://www.transplantmodels.com/esdrisk/>). Additional risk attributable to donation is likely to be 3 to 5 times higher than risk in the absence of donation, but there is substantial uncertainty, especially in younger donor candidates, and we suggest caution in decision making. Postdonation risk above the threshold is justification for a decision to decline. Candidates with risk below the threshold are acceptable to make their own decision whether to donate. Colors are blended together to signify threshold for decision making is imprecise. eGFR, Estimated glomerular filtration rate; eGFRcr, estimated glomerular filtration rate based on serum creatinine; eGFRcr-cys, estimated glomerular filtration rate based on serum creatinine and cystatin; eGFRcys, estimated glomerular filtration rate based on cystatin; ESKF, end-stage kidney failure; GFR, glomerular filtration rate; mCrCl, measured creatinine clearance; mGFR, measured glomerular filtration rate.

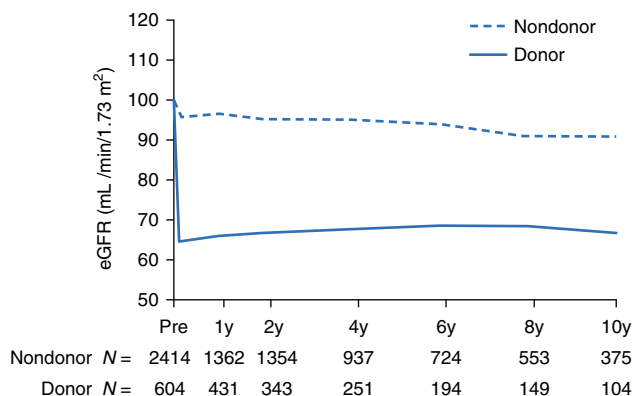


Fig. 70.4 Mean estimated glomerular filtration rate (eGFR) over time in living kidney donors (solid line) and matched healthy nondonor controls (dotted line). (Reproduced with permission.¹⁹)

OBESEITY AND DIABETES

Obesity is a strong risk factor for diabetes, cardiovascular disease, and kidney disease. Living donor nephrectomy is more difficult for patients with excess visceral fat, increasing the risk of perioperative complications including infection, blood loss, and delayed wound healing. Various body mass index (BMI) cut-points have been reported in the literature as absolute or relative contraindications to donation. Elevated serum glucose or glucose intolerance are also strong risk factors for diabetes mellitus. Apart from personal and family history assessments of diabetes mellitus (childhood, adult-onset, gestational), glycosylated hemoglobin and serum and urinary glucose are typically measured early in the assessment of all candidates. Fasting glucose and oral glucose tolerance tests are recommended for high-risk candidates (e.g., high random glucose, positive family history). According

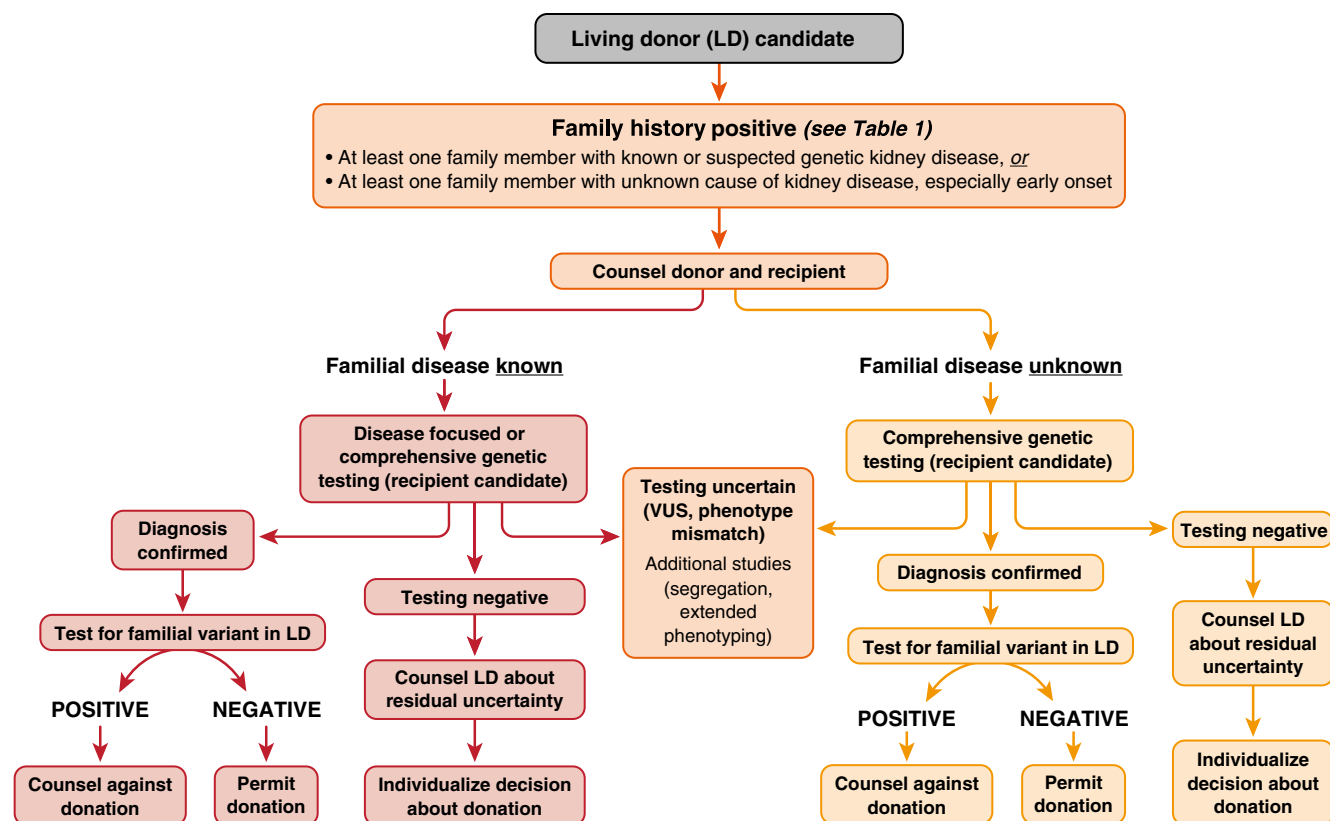


Fig. 70.5 Approach to genetic testing of living donor candidates. LD, living donor; VUS, variant of uncertain significance. (Reproduced with permission.²³)

to the 2017 KDIGO living kidney donor guideline, donor candidates with type 1 diabetes mellitus should not donate.¹¹ The decision to approve donor candidates with prediabetes or type 2 diabetes mellitus should be individualized on the basis of demographic and health profile in relation to the transplant program's acceptance threshold. Donor candidates with prediabetes and type 2 diabetes mellitus should be counseled that their condition may progress over time and may lead to end-organ complications, regardless of whether or not they donate. Less evidence is available to comment on the influence of predonation lipid levels (e.g., cholesterol, triglycerides, and high-density and low-density lipoproteins) and predonation active smoking on donor candidacy, although notably, smoking was a strong risk factor for ESKF in healthy persons.³¹ While candidates should be educated and encouraged to modify their dietary and smoking habits, eligibility based on these factors may vary across programs.

SCREENING FOR TRANSMISSIBLE INFECTIONS

To reduce the risk of viral transmission from the donor to the recipient, the evaluation should include assessment of prior history of infections, recent travel history, and virology screens early in the evaluation and again within the 2 to 4 weeks of donation to minimize the window of infection.³² The 2017 KDIGO living kidney donor guideline recommends screening for human immunodeficiency virus (HIV), hepatitis B and C, Epstein-Barr virus, cytomegalovirus, syphilis, urinary tract infection, and

other potential infections based on geography and environmental exposures.¹¹ If a donor candidate is found to have a potentially transmissible infection, then the donor candidate, the intended recipient, and transplant team should weigh the risks and benefits of proceeding with donation and develop a management plan if the decision is to proceed with donation. One case series in the United States reported three successful LDKT from donors with HIV to transplant recipients with HIV and showed promising results after 2 to 4 years of follow-up, although the long-term postdonation risk remains unknown.³³ In South Africa, ~20% of people aged 15 to 64 years are living with HIV in South Africa; therefore facilitating safe organ donation and transplantation from HIV-positive donors could increase donation and transplantation rates. A clinical practice guideline has been developed for HIV-positive organ donors and recipients, which considers eligibility, risk of transmission, and ethical considerations.³⁴

The coronavirus disease 2019 (COVID-19) pandemic had a global impact on organ donation and transplantation activity, especially LDKTs. From 2010 to 2019, the global rate of LDKT was stable at 4.6 pmp and then dramatically decreased to 3.3 pmp in 2020.⁶ In the early waves of the pandemic, many transplant programs decreased or suspended living donor candidate evaluations and LDKT surgeries for a variety of reasons.^{35,36} This included safety concerns for the donor, recipient, and health care staff; strains on the health care system with many intensive care units converted to COVID-19 units; and hospital and government restrictions.^{35–37} By 2022, the global LDKT

activity had improved to 6.7 pmp⁶ with many centers screening donor and recipient pairs for COVID-19 infection, mostly using the polymerase chain reaction testing from nasopharyngeal swab samples.³⁸ One retrospective cohort study from India reported on 64 LDKTs in which both the donor and the recipient had previously recovered from COVID-19 infections.³⁹ After a median follow-up of 227 days (interquartile range, IQR, 109–309 days), the outcomes were reassuring for both the donor and the recipient with no reported deaths, graft loss, reactivation/reinfection, or complications related to the surgery or COVID-19.³⁹ This suggests that it is safe to proceed with donor candidates who have cleared the infection without residual lung, kidney, or systemic sequelae. There is considerable variability in the practice patterns for accepting living kidney donors with a history of COVID-19 infections, suggesting that further research is needed to better guide optimal practices to ensure the safety of living donation and LDKT as COVID-19 transitions from a pandemic to an endemic infection.⁴⁰

CANCER SCREENING

All candidates should be up to date with local cancer screening guidelines according to age, sex, and family history. Donors with active cancer are generally not eligible to donate. Donors with a prior history of successfully treated cancer with a high risk of recurrence may be excluded from donation because antineoplastic agents may be nephrotoxic and transmission of cancer from the donor to the recipient can have serious consequences for the immunocompromised recipient.⁴¹ Candidates with a prior history of cancer with a minimal or low risk of recurrence (<1%, e.g., nonmelanoma skin cancers, and solitary papillary thyroid carcinomas) may be considered on a case-by-case basis. In some cases, candidates with small renal tumors (high-grade Bosniak renal cysts [III or higher] or small [T1a] renal cell carcinoma curable by nephrectomy) may be acceptable for donation, and the donor and recipient must provide consent for the cancer to be resected at the time of donor nephrectomy.^{42,43}

MATERNAL AND FETAL COMPLICATIONS

During the evaluation process, female donor candidates should be asked about any prior history of pregnancy complications, including gestational diabetes, gestational hypertension, and preeclampsia, as well as any future childbearing plans. Women of childbearing age should not be excluded from donation solely because they wish to have children after donation, but they should be counseled on the potential increased risk of pregnancy complications postdonation (see Risks and Outcomes of Living Kidney Donation). Women with a prior history of pregnancy complications should be informed of their potential increased risk of long-term complications, with or without donation, and any uncertainty in the donor-attributable risks. One single-center study from the United States included 1862 female living kidney donors (1963–2020) with predonation pregnancies.⁴⁴ The median time from first pregnancy to donation was 18.5 years, and the median postdonation follow-up was 18.0 years. Of these donors, 43 had a history of predonation gestational diabetes mellitus, 49 had gestational hypertension, and 48 had preeclampsia/

eclampsia. Compared with matched donor controls, donors with predonation gestational diabetes mellitus had a 3-fold increased risk of developing postdonation diabetes mellitus (hazard ratio [HR] 3.04, 95% confidence interval [CI] 1.33–6.99, $P \leq 0.01$). Those with predonation gestational diabetes mellitus had a 17% chance of developing diabetes mellitus within 20 years after donation compared with an 8% chance for matched controls. Donors with predonation gestational hypertension had a 2-fold increased risk of developing postdonation hypertension (HR 1.89, 95% CI 1.26–2.83, $P \leq 0.01$). Those with predonation gestational hypertension had a 53% chance of developing hypertension within 20 years after donation compared with a 32% chance for controls. Lastly, donors with predonation preeclampsia/eclampsia had a twofold increased risk of developing postdonation cardiovascular disease, although this did not reach statistical significance. While the 2017 KDIGO guideline states that women with a predonation history of pregnancy complications could still potentially donate, other guidelines consider this to be a relative/absolute contraindication to donation.^{11,45}

PSYCHOSOCIAL ASSESSMENT

The 2017 KDIGO living kidney donor guideline recommends that a psychosocial assessment be conducted for all donors (regardless of relationship with the intended recipient), in the absence of the intended recipient (to reliably assess voluntariness), and by a professional independent from the care of the intended recipient.¹¹ The psychosocial assessment should also include evaluation for mental health issues, including drug and alcohol dependence and addiction, available social supports for the donor before and after surgery, and addressing any financial concerns, including health insurance coverage, with available resources to help ensure a financially-neutral assessment. A thorough psychosocial assessment should minimize the incidence of poor psychosocial outcomes postdonation by careful selection and possible predonation treatment (e.g., counseling), and by ensuring adequate support are available, if needed, after donation. Quality of life is generally positive postdonation, but there have been instances of regret, depression, and financial hardships.⁴⁶ In a Canadian qualitative study, living donors identified a lack of psychosocial care beyond screening and evaluation as a significant gap in their care provision.⁴⁷

PAIRED DONATION AND INCOMPATIBLE LIVING DONOR TRANSPLANTS

For incompatible donor-recipient pairs, either ABO- or human leukocyte antigen (HLA)-incompatible, candidates should be informed of the availability, risks, and benefits of kidney paired donation (KPD); incompatibility management strategies; or waiting for a suitable living or deceased donor. The emergence of KPD, together with logistical advances (e.g., coordinating multiway exchanges and shipping organs rather than the donor), has increased the potential for KPD to substantially increase the number of living donor transplants performed.⁴⁸ Some programs even permit compatible donors to participate in KPD

should they choose to potentially match to a living donor who is younger or has fewer donor-specific antibodies.⁴⁹ In the United States, from 2006 to 2021, the number of KPD transplants has increased over time with almost 1 in 5 LDKTs in 2021 performed via the KPD program.⁵⁰ In 2019, a survey of experts in the European Network for Collaboration on Kidney Exchange reported 10 mature operating programs and other programs in preparation.⁵¹ In developing nations, there are challenges in implementing KPD programs, such as system inefficiencies, logistics related to allocation, and reimbursement for direct and indirect expenses to the donor.⁵²

DONOR NEPHRECTOMY APPROACHES

The surgical approach to donor nephrectomy is based on the surgeon's experience, as well as the donor's history, physical examination, renal anatomy and function, and personal preference. Following an uncomplicated donor nephrectomy, most donors can expect to spend 2 to 7 days in hospital and return to work in 4 to 9 weeks.⁵³ Postdonation risks are described later.

LAPAROSCOPIC VERSUS OPEN VERSUS ROBOTIC DONOR NEPHRECTOMY

Laparoscopic donor nephrectomy has replaced the open technique as the standard of care, with the laparoscopic approach accounting for >90% of all living donor nephrectomies performed in the United States.⁵⁴ The conversion rate from laparoscopic to open donor nephrectomy ranges from 1% to 1.8%.⁵⁵ Compared with open nephrectomy, laparoscopic nephrectomy has advantages including less perioperative blood loss, reduced postoperative pain, shorter hospital stay, faster return to work, and better cosmetic appearance.^{53,55} Advances in surgical techniques, including the use of robotic donor nephrectomy, aim to improve these donor experiences and outcomes. One meta-analysis of 17 studies with 6970 donors compared robotic-assisted donor nephrectomy to laparoscopic donor nephrectomy and found that in a subgroup of experienced surgeons, robotic-assisted donor nephrectomy was associated with shorter operating time and length of hospital stay, lower conversion to open nephrectomy, and reduced surgical complications.⁵⁶ This suggests that robotic-assisted nephrectomy could become an increasingly preferred approach for living donor nephrectomy, although adequate training and experience are important to ensure donor safety.

LEFT VERSUS RIGHT DONOR NEPHRECTOMY

The laterality of donor nephrectomy may be influenced by anatomic or functional factors. Multiple renal arteries, either unilaterally or bilaterally, may be technically more complicated or increase the risk of dissection and bleeding. Structural abnormalities, such as the presence of renal cysts or calculi, may result in the removal of the affected kidney, with or without additional bench excision, leaving the donor with the unaffected kidney. Similarly, for donors with significant differences in split kidney function (>55%

vs. <45%), the recommendation is to leave the donor of the kidney with better function.¹¹

Beyond these considerations, left nephrectomy is generally preferred for technical reasons due to the longer left renal vein compared with the right.^{11,57} A systematic review and meta-analysis found that, compared with right-laparoscopic donor nephrectomy, there was less delayed graft function and thrombosis with left-laparoscopic donor nephrectomy.⁵⁸ However, there was no significant difference in other surgical outcomes, including operative times, blood loss, warm ischemia time, and length of stay.

LIVING KIDNEY DONOR FOLLOW-UP

A personalized plan for early- and long-term living kidney donor follow-up care should be discussed during the evaluation process.¹¹ This follow-up care plan may outline the timing and frequency of visits, as well as who will be providing the follow-up care (primary care provider vs. transplant center). Early follow-up care (within 3–6 months) is routinely done, usually by the surgical team, as part of the postoperative care plan.

The 2017 KDIGO living kidney donor guideline recommends lifelong annual assessment of kidney health, including blood pressure measurement, serum creatinine testing for eGFR, and urine samples for albuminuria.¹¹ Currently, it is unclear how frequently living kidney donors are receiving follow-up care. Follow-up care at 1, 5, 10, and 15 years in Norway is achieved by 99%, 95%, 84%, and 77% of donors, respectively.¹¹ In a retrospective Canadian study of 534 living kidney donors from a single province with a median follow-up of 6.5 years (maximum 12.5 years), 25% of donors had all three markers of care (physician visit, serum creatinine, albuminuria testing) performed on an annual basis.⁵⁹ The adherence to physician visits was higher than serum creatinine or albuminuria testing. In a Canadian qualitative study, many donors who were not offered long-term follow-up teams described feelings of abandonment and detachment from the health care system.⁴⁷

In 2013, the United Network for Organ Sharing implemented mandatory reporting of clinical and laboratory data for living kidney donors at 6, 12, and 24 months after donation by the transplant centers to the national transplant registry.⁶⁰ Donor follow-up data beyond 2 years, when medical conditions are more likely to develop, is not currently mandated in the United States. One Canadian study reported that living kidney donors who had early guideline concordant follow-up care in the first 2 years postdonation were more likely to continue to have annual guideline-concordant care; however, there was no significant impact on change in eGFR over time or rates of all-cause hospitalization.⁶¹

Barriers to follow-up for living kidney donors include sociodemographic and geographic factors, perception of unnecessary need for follow-up due to excellent donor health, outdated donor contact information, the cost of follow-up to the donor, and resource and personnel constraints.⁶² Yet there are numerous reasons to ensure adequate long-term follow-up care for living kidney donors. Follow-up care may result in prevention, early detection, and treatment of diseases, particularly those associated

with reduced kidney function and its complications. It also provides an opportunity to document medical and surgical complications that have occurred that may be attributed to donation. This information can be used to develop interventions to mitigate these risks and can also improve the informed consent process for future living kidney donors. Lastly, ongoing follow-up care allows the health care team to update and educate prior donors on relevant outcomes as new research arises.

Recognizing the importance of living donor follow-up, the European Living Donation and Public Health project mandates living donor registration and follow-up data collection through a centralized donor database.⁶³ In the United States, the Health Resources and Services Administration asked the Scientific Registry of Transplant Recipients to establish a national Scientific Registry for Living Donors.⁶⁴ This registry aims to collect follow-up data on living donors, including living donor candidates who undergo the evaluation process but do not end up donating. Through staff contact and novel data linkages, the registry seeks to promote lifelong living donor follow-up. Longitudinal monitoring of medical and psychosocial outcomes important to donors and families should be improved globally, through creation and expansion of registries to track outcomes in all countries performing living donation.

RISKS AND OUTCOMES OF LIVING KIDNEY DONATION

While most living kidney donors will not experience major morbidity or mortality after donation, donation does pose some risk of surgical, medical, and psychosocial complications.^{65,66} Comprehensive data on long-term outcomes for living kidney donors are limited due to single-center studies with small sample sizes, insufficient power to quantify rare events, significant missing data or lost to follow-up, short duration of follow-up, lack of donor diversity in race and comorbidities, and lack of a comparable control group. Living kidney donors are carefully selected following a rigorous evaluation process and as a result are inherently healthier than the general population. Comparing outcomes in donors with nondonors selected for comparable baseline health offers a perspective of risk that can better capture the attributable risk of donation (Table 70.4, Fig. 70.6).^{65,66} In addition to this, the characteristics of living kidney donors have evolved over time. Transplant programs are now accepting donors who are older with more comorbidities including hypertension and obesity. Such practices are defensible if the risks, based on a candidate's demographic and health profile, are acceptably low. Ongoing work is needed to capture and quantify risks in groups with less aggregate follow-up to date, such as ethnic minorities and medically complex donors. It remains valuable to build efficient donor candidate registries with long-term follow-up of donors and nondonor controls.^{64,67}

PERIOPERATIVE MORTALITY AND COMPLICATIONS

In a study of 164,593 living kidney donors from the United States (1993–2022), the 90-day surgical mortality rate was

2.2 per 10,000 donors and much lower in the more recent era (1993–2002 vs. 2003–2012 vs. 2013–2022: 3.0 vs. 2.9 vs. 0.9 per 10,000 donors).⁶⁸ The most common cause of death was hemorrhage. The incidence of perioperative morbidity has been reported to be <20% for minor complications and <5% for major complications.^{69,70} A study integrating national donor registry data from the United States (2008–2012) with administrative records from a consortium of 98 academic hospitals found that 16.8% of living kidney donors ($n = 14,964$) experienced a perioperative complication, most commonly gastrointestinal (4.4%), bleeding (3.0%), respiratory (2.5%), and surgical/anesthesia-related injuries (2.4%).⁶⁹ Major complications were identified in 2.5% of donors. Another multicenter prospective study of 1042 living kidney donors from Canada and Australia (2004–2014) reported that 13% of donors experienced a perioperative complication, most commonly organ injury and ileus.⁷⁰ Major complications occurred in 1% of donors and there were no reported perioperative deaths.

LONG-TERM MORTALITY

A meta-analysis of 19 studies involving 8098 living kidney donors reported that the pooled mortality was 3.8% over 10 years postdonation.⁷¹ The risk of all-cause mortality is lower in living kidney donors compared with the general population, likely due to the rigorous evaluation and selection process. In a meta-analysis of 4 studies with 84,495 donors compared with 62,484 healthy nondonors, there was no increased risk of all-cause mortality in donors (1.7% vs. 5.0%; pooled adjusted risk ratio 0.60, 95% CI 0.31–1.10) over an average follow-up of 6 to 15 years.⁷² In one of the studies from Norway, the authors reported that in the first decade of follow-up, the risk of all-cause mortality was similar in donors ($n = 1901$) compared with healthy matched nondonors ($n = 32,621$); however, at 25 years of follow-up, the cumulative incidence of all-cause mortality was higher in donors compared with nondonors (~18% vs. ~13%; adjusted HR 1.3, 95% CI 1.1–1.5).⁷³ The most common cause of death among living kidney donors was cardiovascular disease, accounting for 30% of all deaths, with an increased risk of cardiovascular mortality in donors compared with healthy matched nondonors (adjusted HR 1.4). Limitations of the Norwegian study include differences in baseline characteristics (e.g., age) and differences in accrual periods between donors and nondonors.^{73,74}

KIDNEY FUNCTION

Despite losing 50% of functioning renal mass, living kidney donors are typically left with 60% to 70% of their predonation GFR due to compensatory hyperfiltration of the remaining kidney. In the first decade after donation, a retrospective study from Canada (605 donors, 2414 healthy nondonors)¹⁹ and a prospective study from the United States (205 donors, 203 healthy nondonors)⁷⁵ both showed that the kidney function in donors is stable with a change in GFR that is significantly different compared with healthy nondonors (donors vs. nondonors: eGFR change of +0.35 vs. –0.85 mL/min/1.73 m² per year in the study from Canada [Fig. 70.4]; mGFR change of +0.02 vs. –1.26 mL/min/1.73 m² per year in the

Table 70.4 Summary of Studies Quantifying Select Long-Term Medical Outcomes in Living Kidney Donors Compared With Healthy Nondonor Controls

Study, Author (year) Country	Living Kidney Donors (n)	Healthy Matched Nondonors (n)	Median Donor Follow-up Time (years)	Donor Age at Donation (years) ^a	Number of Donor Events (%) [Event Rate]	Number of Healthy Nondonor Events (%) [Event Rate]	Event Rate Hazard Ratio (95% Confidence Interval)	P Value
End-Stage Kidney Failure								
Muzaale ⁷⁸ (2014) United States	96,217	96,217	7.6	40 (11)	99 (0.10%) [30.8 per 10,000]	36 (0.037%) [3.9 per 10,000]	NR	$P < 0.001$
Mjøen ⁷³ (2014) Norway	1901	32,621 ^b	15.1	46 (11)	9 (0.47%)	22 (0.067%)	11.38 (4.37-29.63)	$P < 0.001$
Acute Kidney Injury Treated With Dialysis								
Lam ¹⁰⁴ (2012) Canada	2027	20,270	6.6	43 [34-50]	1 (0.05%) [6.5 per 100,000]	14 (0.07%) [9.4 per 100,000]	0.58 (0.08-4.47)	$P = 0.61$
All-Cause Mortality								
Segev ¹⁰⁵ (2010) United States	80,347	9364	6.3	NR	1205 (1.5%)	272 (2.9%)	NR	NR
Berger ¹⁰⁶ (2011) United States (≥70 years old)	219	219	NR	72 (2.3)	22 (10%)	59 (27%)	0.37 (0.21-0.65)	$P < 0.001$
Garg ⁹² (2012) Canada	2028	20,280	6.5	43 [34-50]	16 (0.8%)	365 (1.8%)	NR	NR
Reese ¹⁰⁷ (2014) United States	3368	3368	7.8	59 (NR)	115 (3.4%) [4.9 per 1000]	152 (4.5%) [5.6 per 1000]	0.90 (0.71-1.15)	$P = 0.21$
Mjøen ⁷³ (2014) Norway	1901	32,621	15.1	46 (11)	224 (11.8%)	2,425 (7.4%)	1.30 (1.11-1.52)	$P = 0.001$
Kim ¹⁰⁸ (2020) Korea	1137	4387	10.9	41 (10)	25 (2.2%) [1.9 per 1000]	77 (1.8%) [1.6 per 1000]	0.99 (0.49-1.98)	$P = 0.9$
Cardiovascular Mortality								
Mjøen ⁷³ (2014) Norway	1901	32,621	15.1	46 (11)	68 (3.6%)	688 (2.1%)	1.40 (1.03-1.91)	$P = 0.03$
Death or Cardiovascular Event								
Reese ¹⁰⁷ (2014) United States	1312	1312	NR	59 (NR)	NR	NR	1.02 (0.87-1.20)	$P = 0.70$
Garg ⁹² (2012) Canada	2028	20,280	6.5	43 [34-50]	42 (2.1%) [2.8 per 1000]	652 (3.2%) [4.1 per 1000]	0.66 (0.48-0.90)	$P = 0.008$
Major Cardiovascular Events								
Garg ⁸⁸ (2024) Canada	924	928	7.3	47 (11)	18 (2%) [0.3 per 100]	15 (2%) [0.2 per 100]	1.21 (0.02-70.59)	

Continued on following page

Table 70.4 Summary of Studies Quantifying Select Long-Term Medical Outcomes in Living Kidney Donors Compared With Healthy Nondonor Controls (Continued)

Study, Author (year) Country	Living Kidney Donors (n)	Healthy Matched Nondonors (n)	Median Donor Follow-up Time (years)	Donor Age at Donation (years) ^a	Number of Donor Events (%) [Event Rate]	Number of Healthy Nondonor Events (%) [Event Rate]	Event Rate Hazard Ratio (95% Confidence Interval)	P Value
Garg ⁹² (2012) Canada	2028	20,280	6.5	43 [34-50]	26 (1.3%) [1.7 per 1000]	287 (1.4%) [2.0 per 1000]	0.85 (0.57-1.27)	P = 0.43
Hypertension								
Garg ⁸⁸ (2024) Canada	924	928	7.3	47 (11)	161 (17%) [2.7 per 100]	158 (17%) [2.5 per 100]	1.11 (0.75-1.66)	—
Albuminuria								
Garg ⁸⁸ (2024) Canada	905	904	—	—	132 (15%) [2.4 per 100]	95 (11%) [1.7 per 100]	1.46 (0.97-2.21)	—
Kidney Stones With Surgical Intervention								
Thomas ²⁸ (2013) Canada	2,019	20,190	8.4	43 [34-50]	16 (0.79%) [8.3 per 10,000]	179 (0.89%) [9.7 per 10,000]	1.04 (0.60-1.80)	P = 0.90
Major Gastrointestinal Bleeding								
Thomas ¹⁰⁹ (2014) Canada	2009	20,090	8.4	42 [34-50]	33 (1.6%) [18.5 per 10,000]	253 (1.3%) [14.9 per 10,000]	1.25 (0.87-1.79)	P = 0.24
Skeletal Fractures								
Garg ¹¹⁰ (2012) Canada	2015	20,150	6.6	43 [34-50]	[16.4 per 10,000]	[18.7 per 10,000]	0.87 (0.58-1.31)	P = 0.50
Gout								
Lam ¹¹¹ (2015) Canada	1988	19,880	8.4	43 [35-51]	67 (3.4%) [3.5 per 1000]	390 (2.0%) [2.1 per 1000]	1.6 (1.2-2.1)	P < 0.001
Gestational Hypertension or Preeclampsia								
Garg ⁹⁵ (2015) Canada	85 ^c	510 ^c	11.0	29 [26-32]	15 (11.5%)	38 (4.8%)	2.4 (1.2-5.0) ^d	P = 0.01

^aData presented as mean (standard deviation) or median [interquartile range].^bLiving kidney donors were not matched to healthy nondonors in the comparison for end-stage kidney failure risk.

study from the United States). Thus in living kidney donors, the GFR steadily increases from the immediate postdonation baseline over the first decade compared with the age-related decline in GFR in nondonors. For a discussion of the impact of nephron number/kidney size on long-term postdonation kidney function in kidney donors, see Chapter 20.

END-STAGE KIDNEY FAILURE

Many transplant programs and countries have reported their incidence of ESKF after living donation, typically defined as receipt of chronic dialysis or kidney transplantation.⁷¹ A

meta-analysis of 12 studies involving 108,900 living kidney donors reported that the overall prevalence of ESKF was 0.5% in the 6 months to 5 years postdonation and 1.1% in the 10 years onward.⁷¹ One single-center study from the United States reported the clinical course of 39 living kidney donors who subsequently developed ESKF after a mean of 27 years.⁷⁶ For these donors, their kidney function remained stable postdonation, until they developed a new comorbidity, such as hypertension or diabetes, that led to progressive CKD.

One study reported that the risk of ESKF in living kidney donors was lower than that in the general unscreened

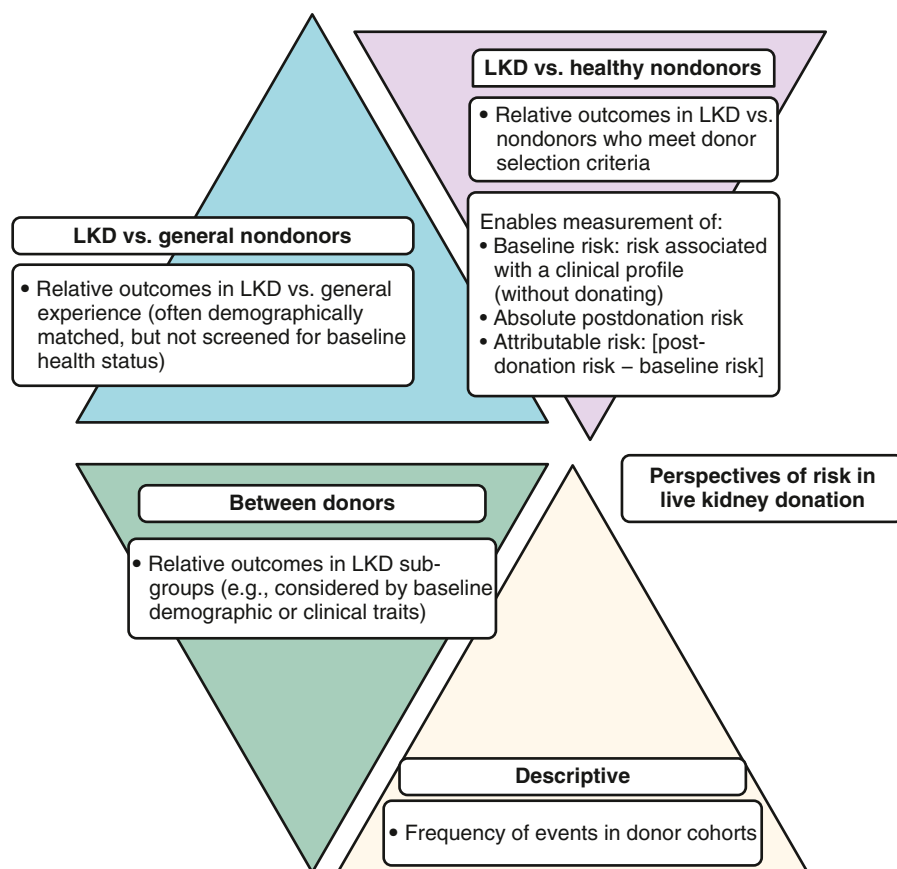


Fig. 70.6 Perspectives of risk in living kidney donation. LKD, Living kidney donor. (From Lentine KL, Segev DL. Understanding and communicating medical risks for living kidney donors: a matter of perspective. *J Am Soc Nephrol*. 2017;28:12–24.)

population (134 vs. 354 cases per million person-years).⁷⁷ In contrast, two studies have reported a higher risk of ESKF in living kidney donors compared with healthy, matched nondonors.^{73,74,78} In the previously described study from Norway, the incidence of ESKF over a median of 15 years was 0.5% ($n = 9$) for living kidney donors and 0.07% ($n = 22$) for healthy, matched nondonors (HR 11.4, 95% CI 4.4–29.6).⁷³ All affected donors were biologically related to their recipient, and the cause of ESKF in all cases was immunologic kidney disease. In another study of 96,217 living kidney donors from the United States, after a median follow-up of 7.6 years (maximum 15.0 years), 99 donors (0.10%) developed ESKF after a mean of 8.6 years after donation.⁷⁸ The 15-year risk of ESKF was higher in donors who were men versus women, Black versus White, and older versus younger (≥ 50 vs. < 50 years). There was no significant difference in the risk of ESKF between donors who were biologically related to their recipient versus those who were unrelated to their recipient. In addition, the risk of ESKF was 86% higher among obese donors compared with nonobese donors (HR 1.86, 95% CI 1.1–3.3), and this risk increased by 7% for each unit increase in donor BMI above 27 kg/m² (HR 1.07, 95% CI 1.0–1.1).⁷⁹ Compared with healthy, matched nondonors, the estimated 15-year risk of ESKD was higher in living kidney donors (30.8 vs. 3.9 per 10,000; risk attributable to donation of 26.9 per 10,000).⁷⁸ The estimated lifetime risk of ESKF was also higher in donors compared with healthy, matched nondonors (90 vs. 14 per 10,000) but lower when compared with the unscreened general population (326

per 10,000). Despite these two studies, which reported a higher relative risk of ESKF in donors compared with healthy, matched nondonors, the absolute risk of ESKF for living kidney donors is low.⁷⁴ Two web-based risk calculators incorporating donor characteristics have been developed to help inform the risk of ESKF for donor candidates (in the absence of donation)³¹ and previous donors⁸⁰ (<http://www.transplantmodels.com/>).

In a qualitative study of living kidney donors who subsequently developed ESKF, most donors did not regret their decision to donate despite the adverse outcome.⁸¹ Interestingly, recipients of kidneys from donors who developed ESKF also had a higher risk of graft loss (74% vs. 56%) and death (61% vs. 46%) at 20 years compared with those who received kidneys from donors who did not develop ESKF, suggesting some intrinsic risk conferred by the donor in the quality of the donated kidney.⁸²

PROTEINURIA

In the general population, heavier proteinuria is associated with an increased risk of CKD, ESKF, cardiovascular disease, and death, independent of the GFR.⁸³ In a meta-analysis of 26 studies with 5337 living kidney donors, the rate of proteinuria increased over time, such that after 10 years postdonation, 18.9% had proteinuria (> 150 mg/day) and 5.7% had overt proteinuria (> 300 mg/day).⁷¹ In a study of 3534 White living kidney donors, the incidence of proteinuria after 18 years of follow-up was 6.1% and

was higher in male versus female donors (HR 1.56) and in donors with an increased predonation BMI (HR 1.10 per kg/m² increment).⁸⁴ Living kidney donors who develop proteinuria should be followed closely to mitigate the risk of cardiovascular and kidney disease progression.

HYPERTENSION

In addition to the rise in blood pressure that occurs with normal aging, the postdonation reduction in kidney function may contribute to further increase in blood pressure (acknowledging there is uncertainty in this assertion given limitations in the methods of currently available studies). A systematic review including 157 living kidney donors reported that after an average of 5 years, the blood pressure was 5 mm Hg higher in donors compared to nondonors (weighted mean for systolic blood pressure 6 mm Hg, weighted mean for diastolic blood pressure 4 mm Hg).⁸⁵ The risk of hypertension may be higher in Black donors compared to White donors⁸⁶ and in obese donors who have a >5% gain in BMI (median follow-up 8 years).⁸⁷ Recently, Garg et al. performed a prospective multi-centered international study of 924 living kidney donors compared to 396 healthy, non-donor controls to determine the risk of new-onset postdonation hypertension, defined as sBP ≥140 mm Hg, dBP ≥90 mm Hg, or treatment with antihypertensive medications.⁸⁸ After a median follow-up of 7 years, there was no significant increase in the risk of hypertension in donors compared with nondonors (17% vs. 17%; 2.7 vs. 2.5 events per 100 person-years; weighted HR, 1.11; 95% CI 0.75–1.66) and no significant increase in sBP above that expected with aging (0.4 vs. 0.5 mm Hg per year). Limitations of the study include that the population was predominately⁸⁹ White (88%), limiting generalizability to other racial and ethnic groups.

To better estimate the risk of postdonation hypertension, data from a single-center in the United States (4055 donors with a median follow-up of 18 years) were used to develop a risk prediction calculator that was validated using data from a single center in Norway (1189 donors with a median follow-up of 9 years).⁹⁰ In the cohort from the United States, 34% of donors reported hypertension at a median time of 16 years postdonation while 29% of donors from Norway reported hypertension at a median time of 17 years postdonation. The investigators have also developed web-based risk prediction calculators for postdonation type 2 diabetes, proteinuria, and reduced kidney function based on the data from 4606 mostly White donors with a median follow-up of 23 years (maximum 56 years) (<https://shiny.biostat.umn.edu/Transplant/KidneyDonor/>).⁹¹ These estimates provide risk predictions for living kidney donors that can be used in the donor selection and informed consent process, as well as guide long-term follow-up care planning.⁸⁰

CARDIOVASCULAR EVENTS

In the general population, reduced GFR, and also proteinuria and hypertension, are associated with cardiovascular events. In living kidney donors, who have undergone a surgical reduction in their GFR, the risk of long-term cardiovascular events may not be increased compared to healthy, matched

nondonors.⁹² In 2028 living kidney donors from Canada (1992–2009), the risk of death-censored cardiovascular events (including myocardial infarction, stroke, and coronary or other revascularization) was similar to 20,280 healthy, matched nondonors (1.7 vs. 2.0 events per 1000 person-years, HR 0.85; median follow-up 7 years, maximum 18 years). However, one study from Norway reported a higher risk of cardiovascular death in donors compared with nondonors (30.4% vs. 28.4%; adjusted HR 1.4, 95% CI 1.0–1.9).⁷³ Cardiovascular risk factors should be monitored postdonation and healthy lifestyles encouraged.

MATERNAL AND FETAL COMPLICATIONS

Compared with predonation pregnancies, postdonation pregnancies may be associated with an increased risk of maternal and fetal complications. One systematic review reported that in 16 studies with 1399 postdonation pregnancies, the absolute risk of preeclampsia increased from a level below that of the general population (~1%–3%) to a level comparable with the general population (~4%–10%), based on 5 studies that assessed the risk of preeclampsia.⁹³ This risk may be attributed to other factors, such as maternal age at the time of pregnancy, rather than to donation. In a single-center study from the United States that followed 384 donors, mostly White (92%), with postdonation pregnancies, donors who experienced preeclampsia/eclampsia ($n = 39$) were at higher risk of developing hypertension (HR 2.70) and diabetes (HR 2.15) compared with matched donors without a preeclampsia/eclampsia pregnancy after a median follow-up of 27 years, although there was no difference in the risk of cardiovascular disease or reduced kidney function (defined as eGFR <45 mL/min/1.73 m²).⁹⁴

In a retrospective study of 85 female donors with 131 pregnancies from Canada (1992–2009), the risk of gestational hypertension and preeclampsia was higher compared with 510 healthy, matched nondonors with 788 pregnancies (11% vs. 5%; odds ratio [OR] 2.4).⁹⁵ In this study, there were no significant differences in preterm births, Caesarean section, postpartum hemorrhage, or low birth rate between donors and nondonors. In contrast, a retrospective cohort study from South Korea compared 56 postdonation pregnancies with 437 propensity-matched nondonors and found no significant differences in gestational hypertension, preeclampsia, or composite outcomes (odds ratio 0.63; 95% CI 0.19–2.14).⁹⁶ While most donors will have uncomplicated pregnancies and deliveries postdonation, it is recommended that pregnant donors have access to adequate perinatal care to reduce the risk of pregnancy complications and potential sequelae.¹¹

ETHICAL, LEGAL, AND POLICY CONSIDERATIONS

More than 100 countries support the Declaration of Istanbul, which prohibits the practice of organ trafficking and transplant tourism, although it is estimated that 5% to 10% of kidney transplants globally are the result of commercial organ transplants.⁹⁷ Local laws and regulations on living donation should be followed and communicated

clearly to donor candidates. Countries and programs should be explicit on their legal mandate to protect vulnerable populations, respect donor candidate autonomy (allowing them to withdraw at any time from the evaluation in a way that is confidential and protected), and prohibit incentives to donation that may create undue influence on the donor candidates' decision to donate. Countries and programs should also be clear on how donor candidates may be identified (e.g., living donor "champions," use of social media to advertise the need for a donor, or independent living donor advocates), as well as the policy on postdonation follow-up. The 2023 international summit "Towards Global Convergence in Transplantation: Sufficiency, Transparency and Oversight" held in Santander, Spain, concluded that living donation should only occur if the protection of the donor, from a clinical and psychosocial point of view, can be guaranteed, and that the development and expansion of living donation programs must occur in parallel with an emphasis on the growth and expansion of deceased donations.⁹⁸

Overall, the practice of LDKT helps bridge the gap between the scarcity of available deceased organs for transplantation. Transplant centers must conduct a thorough and comprehensive evaluation process that leads to the selection of living kidney donors who have minimally acceptable risks postdonation. Clinical initiatives should aim to continually improve the efficiency of the donor evaluation process while minimizing cost and inconvenience to the donor candidate. Understanding donor-attributable risks has improved the informed consent process for donor candidates and also provided education for prior living kidney donors. Future research is needed to address gaps in the current state of knowledge, including outcomes in medically complex donors and donors of racial minorities. The transplant community has an ethical obligation to ensure the safety and integrity of living kidney donation and transplantation.

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QUESTIONS

- Which of the following is NOT a risk of living kidney donation described in the literature?
 - Death in the perioperative period (approximately 1–3 in 10,000 donor nephrectomies)
 - Postdonation gestational hypertension or preeclampsia in pregnancy (approximately 1 in 10 donors will develop outcome compared with 1 in 20 nondonors)
 - A higher risk of fracture (10-year risk: approximately 1 in 300 donors will fracture compared with 1 in 600 nondonors)
 - A higher risk of end-stage kidney disease (40-year risk: 1 in 100 donors will develop kidney failure compared with 1 in 400 nondonors)

Answer: c

Rationale: As with any surgery, there is a perioperative risk of death with living kidney donation. Three studies to date suggest a higher risk of gestational hypertension and preeclampsia in donor pregnancies compared with nondonor pregnancies. Kidney donation results in a decrease in the concentration of serum phosphate and plasma calcitriol and an increase in the concentration of serum parathyroid hormone, with no change in the concentration of serum calcium. However, no study has shown donors have a higher risk of fractures compared with nondonors (see [Table 70.4](#)). Two studies highlight a higher risk of kidney failure in donors compared with nondonors, but the absolute chance of developing this outcome in the 15 years after donation is low (<0.5%).

- Which of the following is a standard test done in most transplant centers for all candidates who are evaluated to become a living kidney donor?
 - Native kidney biopsy
 - Hepatitis B serology
 - 24-urine collection for kidney stone metabolic testing
 - 24-hour ambulatory blood pressure
 - Oral glucose tolerance test

Answer: b

Rationale: To prevent unintended transmission of infection from a living donor to a recipient, the donor candidate has recommended tests for infectious disease screening. Some of these tests are also repeated close to the transplant surgery. The 2017 KDIGO guideline recommends screening for the following viruses: human immunodeficiency virus, hepatitis B and C, Epstein-Barr virus, cytomegalovirus, and syphilis. The regulatory agencies of many countries also have specific guidance around infectious disease testing in living donors. All the other tests listed are done in some (but not all) donor candidates based on their risk factors and initial test results.

- Which of the following are NOT goals of the living kidney donor candidate evaluation?
 - Educate donor candidates about the process and likely outcomes of living kidney donor transplantation.
 - Determine whether the candidate has acceptable medical and psychologic risks of donation to proceed with nephrectomy.
 - Minimize the medical and psychologic risks of donation before nephrectomy for candidates who are proceeding with nephrectomy.
 - Develop a postdonation plan to minimize the short- and long-term medical and psychosocial risks of donation.
 - Ensure the intended recipient maintains a good relationship with the donor candidate after the donation process.

Answer: e

Rationale: The 2017 KDIGO guidelines describe the goals of the evaluation of living kidney donor candidates. This includes understanding a donor candidate's willingness to donate a kidney voluntarily without pressure, assessing the benefits and risks of donation for each donor candidate, supporting candidate decision making through education and counseling, developing a plan for postdonation follow-up and care, and for excluded donor candidates developing a plan for any needed care and support. The evaluation team can provide advice to donors including what might happen to their relationship with the recipient after donation. However, the evaluation team is not able to ensure recipients maintain a good relationship with donors after the donation process.